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ORIGINAL ARTICLE

The social cyborg consumer: posthumanist framework for socio-technical consumer behavior

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Abstract

Contemporary consumers increasingly function as human–machine hybrids, challenging traditional consumer behavior models that assume a single, bounded human decision-maker. This conceptual paper introduces the Social Cyborg Consumer framework, which integrates posthumanist philosophy of technology with mainstream consumer behavior research and behavioral economics to reconceptualize consumer choice as co-produced by human cognition and machine intelligence. Building on classical decision models and digital consumer behavior research, we argue that extant theories often treat AI, algorithms, and platforms as external influences rather than as constituents of the consumer's own decision system. We develop six socio-technical "cyborg dimensions" of consumer behavior—Extended Cognition, Hybrid Agency, Algorithmic Choice Architecture, Networked Self, Data-Driven Personalization, and Attention Governance—each grounded in posthumanist and socio-technical theory (e.g., Haraway, Simondon, Stiegler, Latour, affordance theory) and behavioral concepts (bounded rationality, nudging, social influence, attention limits). Methodologically, we adopt a conceptual research design following MacInnis (2011) and Jaakkola (2020): an interdisciplinary literature review, abductive clustering of recurring themes, iterative refinement into six analytically distinct dimensions, and illustrative mini-cases from sectors such as e-commerce, digital platforms, financial services, and health. We present an integrated conceptual model that locates the dimensions across three layers (cognitive/agency core, social–data layer, and environmental/attention layer) and relates them to consumer autonomy and market ethics. The paper concludes by outlining sector-specific implications for retailers, platforms, financial and health service providers, and regulators, and by specifying avenues for empirical research that can test and extend the Social Cyborg Consumer framework.

Keywords Social cyborg consumer · Posthumanism · Consumer behavior · Digital marketing · Behavioral economics · Extended cognition · Hybrid agency ·

Extended author information available on the last page of the article

Algorithmic choice architecture · Networked self · Personalization · Attention economy

Introduction

Ordering groceries via a smartphone app that pre-populates a cart, accepting a ride suggested by a mobility platform, allowing a robo-advisor to rebalance one's investments, or following a health app's nudges to walk more: these everyday situations illustrate how contemporary consumer decisions are co-produced by human cognition and machine intelligence. Recommender systems, default settings, news feeds, and voice assistants select, filter, and prioritize options before consumers even become aware of them. The result is not simply a "digital consumer" but a socio-technical consumer, whose choices emerge from a tightly coupled human–machine system.

Classic consumer behavior models provide only a partial account of this reality. The Engel–Kollat–Blackwell (EKB) model conceptualizes decision-making as a sequence of stages—problem recognition, information search, evaluation of alternatives, purchase, and post-purchase—performed by an individual decision-maker who gathers and processes information from the environment (Engel et al. 1995). The Theory of Planned Behavior (TPB) similarly explains consumer action through attitudes, subjective norms, and perceived behavioral control located in the individual (Ajzen 1991). Information-processing perspectives (e.g., Bettman 1979) treat the consumer as a bounded but still central processor of stimuli.

These frameworks have been foundational and remain indispensable. However, they typically assume:

1. A single, human locus of cognition and agency, distinct from tools and artifacts.
2. A relatively passive environment, in which technology provides information or channels but does not actively co-decide.
3. Choice architectures designed by humans for humans, with limited adaptive or algorithmic behavior.

By contrast, digital markets are characterized by algorithmic choice environments and AI-mediated interactions (Lamberton and Stephen 2016; Puntoni et al. 2021). E-commerce interfaces curate assortments; social networks algorithmically rank content; smartphone apps track, personalize, and nudge; and autonomous systems can execute purchases with minimal human intervention. While research on digital consumer behavior, online decision environments, and personalization has grown rapidly (e.g., Belk 2013; Tam and Ho 2006; Resnick et al. 2006; Tucker 2014; Hofacker et al. 2016), this work often still treats technology as an external influence or context variable—part of the environment blade of Simon's (1956) "scissors"—rather than as a constitutive part of the consumer's own decision system.

Parallel developments in philosophy of technology and posthumanism have called for rethinking the human–technology relation. Posthumanist authors argue that identities and agencies are distributed across human and non-human elements (Haraway 1985; Braidotti 2013; Hayles 1999). Simondon (2017) and Stiegler (2010, 2014) con-

ceptualize humans and technical systems as co-evolving, while the extended mind thesis (Clark and Chalmers 1998) suggests that tools integrated into our routines become part of our cognitive apparatus. Actor–Network Theory (Latour 2005) and affordance theory (Gibson 1979; Leonardi 2011) similarly highlight socio-technical hybrids and the action possibilities of artifacts.

Taken together, these perspectives suggest that contemporary consumers are best understood as social cyborgs: human–machine hybrids embedded in networks of other humans, devices, platforms, and institutions. Building on earlier calls for post-humanist perspectives in marketing (Campbell et al. 2010), this paper develops a framework to conceptualize such consumers in a way that is both philosophically grounded and actionable for mainstream consumer and marketing research.

We address the following research questions:

- RQ1. How can consumer behavior be conceptualized when human cognition is deeply entangled with machine intelligence across market settings?
- RQ2. Which socio-technical dimensions of this entanglement are most critical for understanding consumer autonomy, welfare, and market dynamics?

To answer these questions, we introduce the Social Cyborg Consumer framework, which articulates six interrelated dimensions of tech-augmented consumer behavior: Extended Cognition, Hybrid Agency, Algorithmic Choice Architecture, Networked Self, Data-Driven Personalization, and Attention Governance. The framework is integrative rather than wholly novel: it synthesizes established ideas from consumer behavior, digital marketing, behavioral economics, and socio-technical theory into a coherent lens tailored to AI-mediated markets.

This article offers a consumer-side socio-technical perspective that complements research on AI, digital transformation, and firm capabilities.

The article proceeds as follows. Section From traditional consumer behavior to socio-technical markets reviews traditional consumer behavior models, digital and algorithmic consumer behavior research, and socio-technical foundations (posthumanism, Actor–Network Theory, affordance theory, and behavioral economics). Section Traditional consumer behavior modelsexplains the conceptual methodology and epistemological stance. Section Digital and algorithmic consumer behavior presents the Social Cyborg Consumer framework: an overview and conceptual model, followed by the six dimensions and their interactions. Section Posthumanism and philosophy of technology develops sector-specific implications for retailers, platforms, financial and health services, and policymakers. Section Actor–network theory and affordance theoryarticulates theoretical contributions and outlines a research agenda. Section Behavioral economics, nudging, and algorithmic decision-making concludes.

From traditional consumer behavior to socio-technical markets

Traditional consumer behavior models

Foundational models such as the Engel–Kollat–Blackwell framework conceptualize consumer decision-making as a sequential process, from problem recognition and information search to choice and post-purchase evaluation (Engel et al. 1995). The consumer is a bounded information processor who collects information from external sources, evaluates alternatives, and selects the best option given constraints (Bettman 1979).

The Theory of Planned Behavior explains behavioral intentions as a function of attitudes, subjective norms, and perceived behavioral control, with behavior largely determined by internal cognitive and motivational states (Ajzen 1991). These and related models have been applied extensively in domains such as retail shopping, environmental behavior, and technology adoption.

However, these frameworks share a human-centered ontology:

- Cognition and agency reside primarily in the individual.
- Technologies are inputs (e.g., information sources, communication channels, product attributes) but not part of the decision-making apparatus itself.
- The structure of the choice environment is largely static and designed ex ante.

This ontology becomes strained when choice environments are dynamically reconfigured by algorithms, when AI systems act as consumer agents, and when consumer identities and preferences are continuously constructed through data flows.

Digital and algorithmic consumer behavior

Research on digital consumer behavior has documented how online environments change information search, evaluation, and decision processes. Studies have examined web personalization and its impact on information processing and choice (Tam and Ho 2006), reputation systems in online marketplaces (Resnick et al. 2006), the role of digital and social media in the customer journey (Lamberton and Stephen 2016; Lemon and Verhoef 2016), and smartphone-mediated experiences (Wang et al. 2016).

The extended self in digital environments (Belk 2013) shows how online possessions, avatars, and profiles become part of consumers' identities, while work on algorithmic consumption and AI in markets shows how algorithms act as market intermediaries and sometimes as consumer agents (André et al. 2018; Smolin et al. 2023). Yet these literatures still tend to treat technology as a contextual factor rather than as part of the consumer's own cognitive and agential system.

Posthumanism and philosophy of technology

Posthumanism challenges the Enlightenment view of the bounded, autonomous subject by treating identity and agency as distributed across biological, technological,

and social systems (Braidotti 2013; Hayles 1999). Haraway's (1985) cyborg figure prefigures contemporary consumers whose memories, perceptions, and actions are extended by devices and code.

Simondon (2017) conceptualizes humans and technical objects as co-individuating: individuals form in relation to associated technical milieus. Stiegler (2010, 2014) argues that human memory and attention are increasingly externalized into technological systems, a process that can both augment and undermine autonomy. Feenberg (2005) highlights the technical code—the way technologies embody social values and power relations—while Zuboff (2019) describes surveillance capitalism, in which data-driven systems monitor and shape behavior.

The extended mind thesis (Clark and Chalmers 1998; Clark 2003) adds a cognitive lens: when artifacts such as notebooks, smartphones, or algorithms are reliably integrated into activity, they become part of the cognitive system. For consumers, search engines, recommender systems, and comparison tools increasingly perform memory and reasoning functions that, if done in the head, would be considered "mental."

Actor–network theory and affordance theory

Actor–Network Theory (ANT) posits that agency arises from networks of human and non-human actors rather than from individuals alone (Latour 2005). A "consumer" is thus a network effect: a node in a web of devices, data, institutions, and other people. From an ANT perspective, a decision to choose a ride-hailing service or a sustainable product is the outcome of interactions among the individual, peers, the platform interface, pricing algorithms, regulatory frameworks, and material infrastructures.

Affordance theory focuses on the action possibilities that technological artifacts offer or constrain (Gibson 1979). Leonardi (2011) extends this to organizational technologies, arguing that affordances arise from the relationship between human goals and material features. In consumer contexts, digital platforms afford particular behaviors—e.g., one-click purchasing, infinite scrolling, easy social sharing—that structure how consumers search, evaluate, and act.

These socio-technical perspectives complement posthumanism by offering more operational concepts (networks, affordances) that can be mobilized in marketing and consumer research.

Behavioral economics, nudging, and algorithmic decision-making

Behavioral economics has long recognized that consumers are boundedly rational (Simon 1956). They satisfice rather than optimize and rely on heuristics shaped by the structure of the environment. Thaler and Sunstein's (2008) notion of choice architecture emphasizes that there is no neutral way to present options; any design will influence choices. Nudge theory recommends designing choice environments that steer people toward beneficial outcomes while preserving freedom of choice (Thaler et al. 2010).

Digital environments amplify these issues. Digital nudging research shows how interface design, defaults, and messages can guide online user behavior (Schneider, Weinmann, & vom Brocke, 2018; Hertwig and Grüne-Yanoff 2017). Algorithmic

systems now not only structure choice environments but also make or strongly recommend decisions—for example, through recommender systems, dynamic pricing, or automated subscription services (André et al. 2018).

In summary, existing consumer and behavioral research offers powerful concepts—bounded rationality, nudging, digital choice environments, social influence, and attention limits (Kahneman 1973)—but has not fully integrated them into a socio-technical conception of the consumer.

To synthesize these interdisciplinary foundations, Table 1 maps each of the six cyborg dimensions to their main philosophical and behavioral origins.

Table 1 Social Cyborg Consumer Dimensions and Key Theoretical Origins

Cyborg Dimension	Philosophical / Socio-technical Origins	Behavioral / Consumer-Research Origins
Extended Cognition	Extended Mind Thesis: mind extends into tools and environments (Clark and Chalmers 1998; Clark 2003). Technics as external memory (Stiegler 2010): technology as cognitive prosthesis.	Bounded rationality and satisficing (Simon 1956). Cognitive offloading and external memory (Sparrow et al. 2011). Information search and decision aids in online shopping (Häubl and Trifts 2000; Yadav and Pavlou 2014).
Hybrid Agency	Cyborg agency (Haraway 1985): hybrid human–machine actors. Human–tech individuation (Simondon, 2017). Algorithmic governmentality (Stiegler 2010). Distributed agency in Actor–Network Theory (Latour 2005).	Self-Determination Theory: autonomy, competence and control (Deci and Ryan 2000). Automation bias and algorithm aversion (Dietvorst et al. 2015). Consumers & AI (Puntoni et al. 2021; André et al. 2018).
Algorithmic Choice Architecture	Technical code (Feenberg 2005): design encodes values and interests. Critical theory of digital architectures (Lessig 2006).	Choice architecture & nudging (Thaler & Sunstein, 2008; Thaler et al. 2010). Digital nudging and interface effects (Schneider, Weinmann, & vom Brocke, 2018; Hertwig and Grüne-Yanoff 2017).
Networked Self	Posthuman identity: decentered, relational selves (Hayles 1999; Braidotti 2013). Actor–Network Theory: persons as network nodes (Latour 2005).	Extended self and digital possessions (Belk 2013). Social influence and herding (Cialdini 2009; Cialdini and Goldstein 2004). Network externalities in adoption (Katz and Shapiro 1985). E-WOM and online reputation (Resnick et al. 2006).
Data-Driven Personalization	Profiling and disindividuation (Stiegler 2010). Infosphere and datafication (Floridi 2014). Surveillance capitalism (Zuboff 2019).	Personalization–privacy trade-offs and targeting (Tam and Ho 2006; Tucker 2014). Analytics and big data in marketing (Hofacker et al. 2016; Wedel and Kannan 2016).
Attention Governance	Attention economy (Stiegler 2014): hyperindustrial capture of attention. Critical accounts of addictive design (Deleuze 1992; Ellul 1964).	Attention as scarce cognitive resource (Kahneman 1973; Simon 1971). Media multitasking and attention fragmentation (Yeykelis et al. 2014). Effects of information overload and engagement maximization on decision quality.

Conceptual methodology and epistemological stance

This paper follows the tradition of conceptual research in marketing (MacInnis 2011; Jaakkola 2020), aiming to develop new integrative constructs and frameworks rather than to test hypotheses empirically.

Epistemological stance

Our epistemology is posthumanist and constructivist. We assume that consumer reality is co-constructed by humans and technologies; neither can be understood in isolation. This implies:

- Treating human and non-human entities symmetrically in analysis (Latour 2005);
- Scrutinizing the power and value structures embedded in technologies (Feenberg 2005; Zuboff 2019);
- Recognizing that cognitive processes extend into material and digital artifacts (Clark and Chalmers 1998; Clark 2003).

Methodologically, this stance means that we use philosophical and socio-technical theories as lenses to reinterpret consumer phenomena, and we treat conceptual coherence, explanatory power, and relevance as key validity criteria (Von Krogh, Rossi-Lamastra, & Haefliger, 2012).

Conceptual research design

Following MacInnis (2011) and Jaakkola (2020), the research design comprised five steps:

1. Phenomenon scoping. We started from the observation that many contemporary consumption episodes (e.g., subscription services, algorithmic feeds, smart devices, robo-advisors) involve close human–machine coupling and delegated decision-making.
2. Interdisciplinary literature integration. We reviewed literatures in:

- Consumer behavior and digital/algorithmic marketing (e.g., Belk 2013; Hofacker et al. 2016; Lamberton and Stephen 2016; Lemon and Verhoef 2016);
- Philosophy of technology and posthumanism (Haraway 1985; Simondon, 2017; Stiegler 2010, 2014; Feenberg 2005; Braidotti 2013);
- Socio-technical theories (Latour 2005; Gibson 1979; Leonardi 2011); and.
- Behavioral economics and choice architecture (Simon 1956; Thaler & Sunstein, 2008; Schneider et al. 2018).

1. Abductive clustering of themes. Through iterative reading and comparison, we identified recurring themes that cut across disciplines, such as cognitive offloading, delegation to algorithms, digital choice architectures, networked identity,

datafication and personalization, and attention capture. These themes were clustered into candidate dimensions.
2. Construct definition and refinement. We consolidated overlapping clusters and sharpened boundaries to arrive at six analytically distinct yet interrelated dimensions: Extended Cognition, Hybrid Agency, Algorithmic Choice Architecture, Networked Self, Data-Driven Personalization, and Attention Governance. For each dimension we developed a definition, theoretical basis, and illustrative market manifestations.
3. Illustrative mini-cases and tabular summarization. We selected well-documented market practices from multiple sectors (e-commerce, mobility, entertainment, financial services, and health) as illustrative cases rather than data sources, using them to demonstrate the plausibility and relevance of each dimension. We then synthesized the framework in tables that link dimensions to theoretical origins, manifestations, and implications.

Table 2 illustrates how each of the six cyborg dimensions manifests in consumer behavior across typical market situations.

Why a conceptual rather than empirical design?

The Social Cyborg Consumer is proposed as a conceptual scaffold for future empirical work rather than as a fully operationalized construct. There are three reasons for this choice:

1. Emergent phenomenon. Many AI-mediated market practices and regulatory responses are rapidly evolving. A conceptual lens is needed first to structure observations and identify key constructs before measurement can be designed.
2. Interdisciplinary integration. Bringing together posthumanist philosophy, sociotechnical theory, and mainstream consumer behavior requires theoretical synthesis that cannot be reduced to immediate hypothesis testing.
3. Research agenda setting. Conceptual frameworks can generate families of testable propositions (Jaakkola 2020). In Sect. Actor–network theory and affordance theory we outline how future research can empirically operationalize and test the proposed dimensions.

Section takeaway. Our approach is intentionally conceptual, grounded in a posthumanist epistemology and an abductive, interdisciplinary research design. The goal is to provide a rigorous yet flexible framework that can guide subsequent empirical studies on machine-augmented consumer behavior.

Table 2 Manifestations of Cyborg Dimensions in Consumer Behavior

Cyborg Dimension	Manifestations in Consumer Behavior (Illustrative Examples)
Extended Cognition	Reliance on smartphones, search engines and voice assistants as primary sources of product and destination information (e.g., "Hey Google, which TV is best?").
	Use of online comparison sites and review aggregators to evaluate quality and price; consumers adopt star ratings as part of their own evaluation heuristics.
	Use of alerts (price-drop notifications, subscription reminders) to outsource continuous monitoring of deals or renewals to digital systems.
Hybrid Agency	Auto-replenishment and subscription services that automatically reorder products within consumer-defined parameters ("subscribe & save").
	Robo-advisors that construct and rebalance investment portfolios based on a consumer's risk profile and goals, with limited ongoing human intervention.
	AI assistants and chatbots that select routes, travel bundles or service options on behalf of users, given high-level preferences and constraints.
Algorithmic Choice Architecture	Personalized rankings and filters on OTAs, marketplaces and search platforms that order options by proprietary "relevance" scores.
	Defaults and pre-selected options online (e.g., add-on insurance, shipping options, tip percentages) that strongly influence final choices.
	Use of dark patterns such as confusing opt-outs, fake urgency timers or hidden fees that exploit cognitive biases to steer behavior.
Networked Self	Purchase decisions driven by social media trends and influencer content (e.g., "TikTok made me buy it"), especially in fashion, beauty and leisure.
	Reputation and rating systems on peer-to-peer platforms (e.g., Airbnb, Uber) where star ratings shape access, matching and prices.
	Participation in brand and lifestyle communities (gaming, fitness, eco-lifestyle, fandoms) that define what is appropriate or desirable to consume.
Data-Driven Personalization	Retargeted advertising and personalized recommendations based on browsing history, purchase history and look-alike audience models.
	Dynamic homepages, emails and app screens that show different content and offers to each user depending on their profile and context.
	Predictive suggestions in online grocery or streaming services (e.g., "you are probably running low on…", "next up") that shape future consideration sets.
Attention Governance	Infinite-scroll feeds and autoplay features that encourage extended usage beyond initial intentions (a quick check-in becomes a long scrolling session).
	Frequent push notifications and alerts that pull consumers back into apps or platforms, even when they are not in an active "shopping" mindset.
	Engagement-optimizing algorithms that prioritize content predicted to maximize watch time, click-through or dwell time, often favouring highly salient items.

The social cyborg consumer framework

Overview and conceptual model

The Social Cyborg Consumer refers to a consumer whose cognition, agency, identity, and attention are deeply entangled with digital technologies and socio-technical networks. These dynamics become particularly consequential under conditions of low-latency human–machine coupling, where interaction unfolds within the tempo of action. In this article, latency is treated as a contextual and intensifying condition of sociotechnical coupling, rather than as a defining variable of the consumer. At the same time, this perspective suggests the need for further research examining how latency may operate as a structuring condition of action in organizational and institu-

tional settings. We conceptualize this configuration through six dimensions organized into three layers. :

1. Cognitive and agency core.

- Extended Cognition: the offloading and extension of cognitive processes into technological tools.
- Hybrid Agency: shared or delegated decision-making between consumers and AI systems.

1. Social and data layer.

- Networked Self: consumer identity and influence emerging from socio-technical networks (peers, influencers, reputational systems).
- Data-Driven Personalization: representation of consumers as data profiles that drive tailored experiences and offers.

1. Environmental and attention layer.

- Algorithmic Choice Architecture: algorithmically configured choice environments that structure and nudge decisions.
- Attention Governance: technological management and commodification of consumers' limited attention.

Extended Cognition and Hybrid Agency can enhance or erode autonomy depending on alignment with consumer goals; Networked Self and Data-Driven Personalization shape identity and fairness; Algorithmic Choice Architecture and Attention Governance structure the possibilities for reflection and choice.

The model can be drawn as concentric layers: a core "human–AI decision system" (Extended Cognition, Hybrid Agency); a middle ring labeled "social/data milieu" (Networked Self, Data-Driven Personalization); and an outer ring labeled "choice environment and attention" (Algorithmic Choice Architecture, Attention Governance). Arrows connect the layers and point toward an outcome box labeled "consumer autonomy, welfare, and market dynamics."

Figure 1 presents the Social Cyborg Consumer framework as a system of concentric layers, integrating the six dimensions, their interactions, and their implications for consumer outcomes.

Extended cognition

Definition Extended Cognition refers to consumers' use of technological tools and systems as integral components of their cognitive processes, such that memory, search, evaluation, and monitoring are performed by human–machine ensembles rather than by the individual alone.

Theoretical basis. This dimension builds on the extended mind thesis (Clark and Chalmers 1998; Clark 2003), Stiegler's notion of technics as external memory

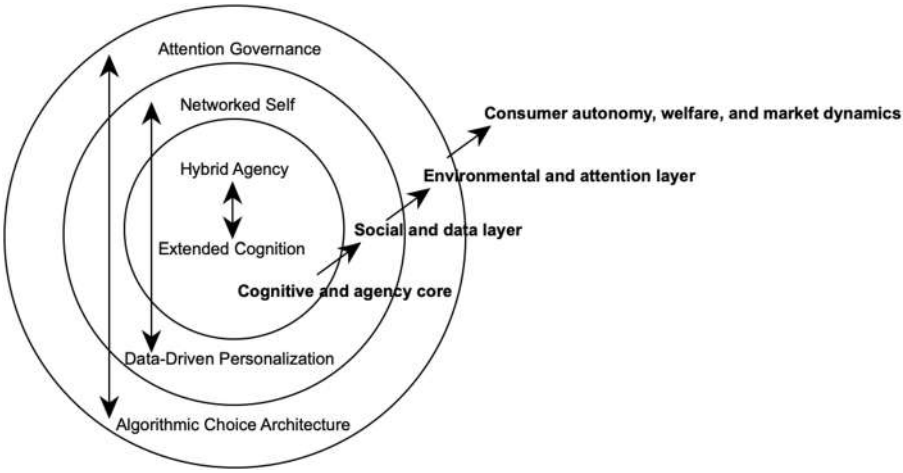


Fig. 1 The Social cyborg consumer: integrated socio-technical framework

(2010), and bounded rationality (Simon 1956). Consumers offload tasks to external devices to cope with cognitive limits, turning search engines, comparison tools, and alerts into parts of their "mind."

Examples.

- Information search. Consumers rely on search engines or voice assistants (e.g., "Hey Google, find the best refrigerator under €800") to retrieve and organize product information, effectively outsourcing knowledge retrieval.
- Decision support tools. Price-comparison websites or loan calculators perform complex computations and present simplified recommendations.
- Monitoring and alerts. Price-drop notifications, stock alerts, and subscription reminders extend consumers' vigilance, allowing "always-on" monitoring without constant attention.

Implications.

For markets, extended cognition can reduce information asymmetry and increase competition when tools are consumer-aligned (e.g., unbiased comparison sites). It can also reallocate power to digital intermediaries that control these tools. For autonomy, extended cognition can be empowering when tools enhance consumers' capabilities, but it risks deskilling and manipulation if tools are biased or opaque (Stiegler 2014). For marketers, the key is to design decision support that is transparent, accurate, and aligned with consumer interests, and to recognize that marketing increasingly targets not only humans but also their cognitive extensions (e.g., how products are represented in comparison algorithms).

Hybrid agency

Definition Hybrid Agency captures situations in which decision-making authority is shared or delegated between consumers and AI systems, such that choices emerge from their interaction rather than from either alone.

Theoretical basis. Drawing on Haraway's cyborg (1985), Simondon's human–tech individuation (2017), and Stiegler's concerns about algorithmic governmentality (2010), Hybrid Agency conceptualizes agency as distributed across human and machine. Actor–Network Theory's notion of distributed agency (Latour 2005) and Self-Determination Theory (SDT) needs for autonomy and competence (André et al. 2018) further inform this dimension.

Examples.

- Auto-restock and subscriptions. Smart appliances or subscription services reorder products automatically within user-defined parameters.
- Algorithmic assistants. Virtual assistants choose restaurants, travel itineraries, or financial products based on user preferences and constraints.
- Robo-advisors. Investment platforms allocate assets according to risk profiles, with varying degrees of human oversight.

Implications.

Hybrid Agency can empower consumers when algorithms act as unbiased, competent agents aligned with consumer goals, strengthening bargaining power against firms (Smolin et al. 2023). However, when algorithms are provider-controlled or conflicted, hybrid agency can subtly shift power away from consumers. Autonomy depends on who sets goals, how transparent the system is, and whether human override is easy. For marketers, Hybrid Agency implies the need to design services with meaningful human-in-the-loop features, clear explanations, and customizable levels of automation.

Algorithmic choice architecture

Definition Algorithmic Choice Architecture refers to the algorithmically configured structure of choice environments—including defaults, rankings, framing, and interface elements—that shape consumer decisions in digital contexts.

Theoretical basis. This dimension extends choice architecture and nudging (Thaler & Sunstein, 2008; Thaler et al. 2010) into algorithmic environments, informed by Feenberg's (2005) technical code and Lessig's (2006) claim that "code is law." Digital nudging research (Schneider et al. 2018) and work on dark patterns provide empirical grounding.

Examples.

- Personalized rankings and filters. Retail platforms sort products by "recommended for you," mixing relevance, profitability, and availability.
- Defaults and pre-selected options. Pre-checked boxes for newsletters, insurance add-ons, or upgrading to premium tiers.

- Dark patterns. Confusing opt-out mechanisms, countdown timers, or misleading color schemes that exploit biases such as scarcity and loss aversion.

Implications.

Algorithmic choice architectures can increase efficiency and facilitate search, but they also enable fine-grained behavioral steering at scale. Ethical concerns arise when architectures exploit cognitive biases without transparency or when platform interests trump consumer welfare. Responsible design requires nudges aligned with users' own goals (Hertwig and Grüne-Yanoff 2017), avoidance of dark patterns, and clear labeling of sponsored content and defaults.

Networked self

Definition The Networked Self dimension captures how consumer identity, preferences, and decisions are shaped by participation in socio-technical networks, including peers, influencers, online communities, and reputational systems.

Theoretical basis. It combines posthumanist accounts of distributed identity (Braidotti 2013; Hayles 1999) with Actor–Network Theory (Latour 2005) and consumer research on social influence and the extended self (Belk 2013; Cialdini 2009). Network externalities (Katz and Shapiro 1985) and electronic word-of-mouth research (e.g., Resnick et al. 2006) provide behavioral foundations.

Examples.

- Social media influence. Trending products on TikTok or Instagram become aspirational consumption norms.
- Reputation and rating systems. Star ratings on platforms such as Airbnb or Uber create digital reputations that influence access and prices.
- Online communities. Brand or lifestyle communities (e.g., eco-friendly living groups) shape members' consumption norms and product repertoires.

Implications.

The Networked Self amplifies social proof and can produce winner-takes-all dynamics. Brands can benefit from cultivated communities but may also face collective backlash (e.g., viral boycotts). From an autonomy perspective, networked influence can both support identity projects and pressure consumers into conformity. Marketers should foster authentic community engagement and avoid deceptive practices such as fake reviews or astroturfing.

Data-driven personalization

Definition Data-Driven Personalization refers to the representation of consumers as granular data profiles that drive tailored content, offers, prices, and experiences, creating feedback loops between behavior, data, and marketing actions.

Theoretical basis. This dimension draws on Stiegler's (2010) concerns about profiling and disindividuation, Floridi's (2014) notion of the infosphere, and marketing research on personalization–privacy trade-offs (Tam and Ho 2006; Tussyadiah and

Miller 2019). It is also informed by work on surveillance capitalism (Zuboff 2019) and targeting effectiveness (Tucker 2014).

Examples.

- Targeted advertising. Ads on social media and search engines are selected based on prior behavior, inferred interests, and look-alike audiences.
- Dynamic content and offers. Retail websites and emails display individualized assortments and promotions; streaming services generate personalized playlists.
- Predictive shopping. Grocery apps suggest likely replenishment items; platforms experiment with anticipatory shipping based on predicted orders.

Implications.

Data-driven personalization promises higher relevance and reduced search costs but raises issues of privacy, fairness, and autonomy. Personalized pricing or differential access can undermine perceptions of equity. Excessive or opaque profiling can feel "creepy" and trigger reactance. Responsible personalization requires permission-based data strategies, transparency ("you are seeing this because…"), and algorithmic fairness checks.

Attention governance

Definition Attention Governance denotes the ways in which technological systems capture, direct, and monetize consumer attention, treating it as a scarce resource to be optimized.

Theoretical basis. Building on Simon's (1973) notion of attention scarcity, Stiegler's analysis of the attention economy (2014), and critical accounts of addictive design (Deleuze 1992; Ellul 1964), this dimension emphasizes the temporal and cognitive aspects of consumption. Media multitasking and attention research (Kahneman 1973; Yeykelis et al. 2014) provide empirical grounding.

Examples.

- Infinite scroll and autoplay. Platforms such as TikTok or Netflix design feeds and playback to keep users engaged for long periods.
- Notification systems. Frequent push notifications and email alerts pull consumers back into apps and sites.
- Engagement-based algorithms. Content ranking systems prioritize items predicted to maximize watch time, click-through, or dwell time.

Implications.

Attention governance has profound implications for well-being and autonomy. Excessive capture can lead to attention fragmentation, impulsive decisions, and digital fatigue. Marketing strategies that prioritize short-term engagement may undermine long-term trust. A more responsible approach is to optimize for quality of engagement (e.g., satisfaction, goal completion) and to provide tools for users to manage their attention (e.g., quiet hours, usage reminders).

Interactions among dimensions

The six dimensions rarely operate in isolation. For example:

- A consumer using a smartphone shopping app engages in Extended Cognition (search and comparison), within an Algorithmic Choice Architecture (ranked lists, defaults), under strong Attention Governance (notifications), while being influenced by their Networked Self (ratings and reviews) and receiving Data-Driven Personalized recommendations. Hybrid Agency emerges when the app's recommendations or auto-reorder features partially decide on behalf of the consumer.
- In health or finance, Extended Cognition and Hybrid Agency are particularly salient (e.g., symptom checkers, robo-advisors), while Data-Driven Personalization and Attention Governance raise heightened ethical stakes due to the sensitivity of the decisions.

Understanding the interplay of dimensions is thus central: the Social Cyborg Consumer is not a sum of separate effects but a socio-technical assemblage in which cognition, data, networks, and attention are tightly coupled.

Sector-specific implications for responsible market design

To address reviewers' call for more concrete and sector-specific implications, this section translates the framework into guidance for four domains: (1) retail and e-commerce, (2) digital platforms and social media, (3) financial and health services, and (4) policymakers and regulators.

Table 3 summarizes the key managerial implications of each dimension for responsible market design and strategy.

Retail and e-commerce

Extended Cognition. Retailers can provide decision aids such as comparison tools, AR try-ons, and educational content to help consumers make informed choices. These tools should clearly disclose how recommendations are generated and avoid hiding important information (e.g., fees) in ways that exploit offloaded cognition.

Hybrid Agency. Subscription and auto-replenishment models should offer granular control (e.g., spending caps, easy pause/stop) and clear reminders before renewals. This preserves the sense that the consumer remains the ultimate decision-maker.

Algorithmic Choice Architecture. E-commerce interfaces should be audited for dark patterns and designed to align with users' stated goals (e.g., filters for sustainable products that are easy to use). "Recommended" labels should be backed by clear, non-manipulative criteria.

Networked Self. Retailers can cultivate authentic customer communities and encourage credible reviews and user-generated content, while actively combating fake reviews and deceptive endorsements.

Table 3 Implications for Responsible Market Design by Dimension

Cyborg Dimension	Responsible Market Design & Strategy Guidelines
Extended Cognition	Design decision-support tools (comparison tables, calculators, AR try-ons) that genuinely clarify options and total costs, rather than obscuring them.
	Provide transparent explanations for recommendations and rankings ("best for you because…"), so that external cognition is not a black box.
	Avoid exploiting cognitive offloading to hide important terms, fees or risks in complex flows; simplify where possible in the consumer's interest.
Hybrid Agency	Offer adjustable levels of automation, from suggestions to full auto-execution, always with a simple override ("undo", "pause", "cancel") option.
	Keep a human in the loop for high-stakes decisions (e.g., major financial or health-related actions), with clear escalation paths to human advisors.
	Make responsibility for algorithmic actions explicit: document decision logic, provide explanations, and offer clear channels for contesting automated decisions.
Algorithmic Choice Architecture	Eliminate dark patterns; regularly audit interfaces for deceptive or manipulative elements (hidden opt-outs, fake scarcity, confusing labels) and remove them.
	Align nudges and default settings with users' stated goals (e.g., savings, health, sustainability) or clear public-interest objectives, not only short-term revenue.
	Ensure opt-out paths are simple, visible and reversible: a neutral "No, thanks" should be as easy to select as the "Yes" option.
Networked Self	Cultivate authentic consumer communities (forums, social groups, UGC spaces) where brands listen and support rather than dominate the conversation.
	Establish and enforce clear influencer-ethics policies: require transparent disclosure of sponsored content and avoid fake followers or inauthentic engagement.
	Monitor online reputation and social conversations; respond quickly and honestly to issues raised in public, and recognize valuable community contributions.
Data-Driven Personalization	Adopt privacy-by-design and permission-based data practices: obtain explicit, understandable consent and communicate the value exchange clearly.
	Provide users with accessible preference centers to view, edit and reset their profiles, and to control the intensity and scope of personalization.
	Audit algorithms for fairness and bias to ensure personalization does not systematically disadvantage specific groups or exploit vulnerabilities.
Attention Governance	Optimize for quality of engagement (satisfaction, goal achievement, trust) rather than maximizing raw screen-time or click volume.
	Limit frequency and intrusiveness of commercial notifications; offer "quiet modes" or "well-being modes" that reduce interruptions by default.
	Ensure clear labelling of sponsored content and avoid purely sensational or manipulative tactics that may harm long-term trust or user well-being.

Data-Driven Personalization. Personalization strategies should rely on transparent consent, allow users to adjust preferences, and avoid discriminatory pricing or offers. Explaining why particular products are recommended can build trust.

Attention Governance. Retailers should avoid excessive promotional spam and time-limited offers that create constant urgency. Fewer, more relevant messages can increase long-term engagement and brand equity.

Digital platforms and social media

Platforms are central arenas of cyborg consumption and thus bear particular responsibility.

- Extended Cognition & Hybrid Agency. Platforms can offer tools that help users understand and manage algorithmic feeds (e.g., "show me more/less of this"), supporting meaningful co-agency.
- Networked Self. Governance of influencers, communities, and reputational systems should prioritize authenticity and protect users from harassment and manipulative content.
- Data-Driven Personalization. Privacy-by-design should be implemented, with granular controls over data sharing and profiling. Users should have access to their data profiles and the ability to correct them.
- Algorithmic Choice Architecture & Attention Governance. Ranking algorithms should be auditable for bias and aligned not only with engagement but also with user welfare (e.g., avoiding promotion of harmful content). Platforms might offer "well-being modes" that reduce addictive features and highlight diverse viewpoints.

Financial and health services

In financial and health domains, the stakes of Hybrid Agency and Data-Driven Personalization are high.

- Hybrid Agency. Roboadvisors and health recommender systems should implement human-in-the-loop safeguards, including mandatory human review for high-impact decisions and clear explanations of recommendations.
- Extended Cognition. Tools that help consumers understand risks, trade-offs, and long-term consequences (e.g., simulations for retirement savings, treatment options) can augment competence and autonomy.
- Data-Driven Personalization. Because financial and health data are highly sensitive, strict privacy protections and explicit consent are essential. Personalization must avoid discriminatory outcomes and comply with fairness and non-discrimination norms.
- Attention Governance. Providers should refrain from exploiting moments of vulnerability (e.g., late-night push offers for high-risk products) and instead design communications to support reflective decision-making.

Policymakers and regulators

The Social Cyborg Consumer framework also suggests priorities for regulation.

- Transparency and explainability. Regulators may require firms to disclose key aspects of algorithmic choice architectures and personalization, especially where significant economic or health outcomes are involved.
- Dark-pattern regulation. Legal definitions and enforcement mechanisms can be developed to curb manipulative interface designs that undermine autonomy.
- Data rights. Rights to access, portability, correction, and deletion of personal data support consumer control over the data-driven layer of their cyborg identity.
- Attention protections. Particularly for minors and vulnerable groups, regulators

might limit highly addictive design practices, mirroring restrictions on other harmful products.

Section takeaway. Seen through the Social Cyborg Consumer lens, responsible market design is about managing human–machine symbiosis: empowering consumers' extended minds and networks while constraining exploitative uses of algorithms, data, and attention.

Theoretical contributions and future research

Positioning and contributions

This paper's contribution is integrative and synthesizing rather than claiming entirely new constructs. It connects and extends four major bodies of work:

1. Traditional consumer behavior and digital marketing. We preserve insights from EKB, TPB, and information-processing models while embedding them in a socio-technical ontology suitable for AI-mediated markets (Engel et al. 1995; Ajzen 1991; Bettman 1979; Lamberton and Stephen 2016; Lemon and Verhoef 2016).
2. Posthumanist and socio-technical theory. By translating posthumanism, extended mind theory, Actor–Network Theory, and affordance theory into six concrete dimensions, we make these often abstract perspectives more accessible and applicable to mainstream marketing scholars (Haraway 1985; Clark and Chalmers 1998; Latour 2005; Leonardi 2011).
3. Behavioral economics and digital nudging. We show how bounded rationality, choice architecture, nudging, and attention limits play out in algorithmic environments, highlighting the ethical responsibilities of firms designing digital choice architectures (Simon 1956; Thaler & Sunstein, 2008; Schneider et al. 2018; Hertwig and Grüne-Yanoff 2017).
4. Digital identity, personalization, and surveillance capitalism. We integrate work on the extended self, personalization–privacy trade-offs, and surveillance capitalism into a coherent story about data-driven, networked, attention-governed consumers (Belk 2013; Tam and Ho 2006; Tucker 2014; Zuboff 2019).

The six dimensions provide a common vocabulary and analytical structure for studying how AI and digital technologies reshape consumer cognition, agency, identity, and attention. They also foreground consumer autonomy and welfare as central evaluative criteria.

Future research agenda

The framework suggests several promising lines for empirical and conceptual research:

1. Operationalization and measurement. Develop valid measures of each dimension—for example, scales for perceived Hybrid Agency (degree of delegation to AI), Extended Cognition (extent of cognitive offloading), or Attention Governance (perceived attention capture).
2. Autonomy and well-being outcomes. Test how combinations of dimensions affect perceived autonomy, satisfaction, and well-being. For instance, does high Extended Cognition coupled with transparent Hybrid Agency enhance autonomy, while opaque Algorithmic Choice Architectures and aggressive Attention Governance undermine it?
3. Consumer heterogeneity. Examine how traits (e.g., need for control, technological literacy, cultural values) moderate responses to cyborg configurations. Some segments may welcome high delegation to AI; others may resist it (Dietvorst et al. 2015; Puntoni et al. 2021).
4. Cross-sector and cross-cultural comparisons. Investigate how the six dimensions manifest differently in sectors such as tourism, mobility, education, and in different institutional environments (e.g., stricter vs. looser data regulations).
5. Design and policy interventions. Conduct experiments on alternative choice architectures, personalization policies, and attention-management features to identify designs that balance firm objectives with consumer autonomy and welfare.

Section takeaway. The Social Cyborg Consumer framework offers a scaffold for a broad research program that can bridge conceptual, empirical, and policy-oriented work on AI-mediated markets.

Conclusion

Consumers in contemporary markets are no longer best described as isolated decision-makers interacting with passive environments. Instead, they are social cyborgs: human–machine hybrids whose cognition, agency, identity, and attention are distributed across devices, algorithms, data, and networks.

This paper has proposed the Social Cyborg Consumer framework, detailing six dimensions—Extended Cognition, Hybrid Agency, Algorithmic Choice Architecture, Networked Self, Data-Driven Personalization, and Attention Governance—that together capture key socio-technical aspects of consumer behavior. Grounded in posthumanist philosophy, socio-technical theory, and behavioral economics, and connected explicitly to mainstream consumer behavior research, the framework responds to calls for more integrated, accessible, and practically relevant theory on AI-mediated consumption.

For marketing scholars, the framework encourages re-examining familiar constructs (e.g., attitudes, decision stages, customer journeys) through a socio-technical lens and provides a vocabulary for studying new phenomena such as consumer–AI collaboration, algorithmic gatekeeping, and attention-aware marketing. For practitioners, it highlights the dual role of technology as both cognitive prosthesis and potential source of manipulation, calling for responsible design that supports, rather than undermines, consumer autonomy and well-being.

Ultimately, recognizing consumers as social cyborgs suggests that marketing in the age of AI is not only about managing customer relationships but also about managing human–machine symbiosis in markets. Firms and policymakers that embrace this perspective will be better positioned to design socio-technical systems that create sustainable value for both businesses and consumers.

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Declarations

Competing interests The author declare no competing interests.

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